

smaller spaces. Much of the interest in such stable splittings comes from the fact that they provide a geometric explanation for algebraic splittings of homology and cohomology groups, as well as other algebraic invariants of spaces that are unaffected by suspension such as the cohomology operations studied in §4.L.

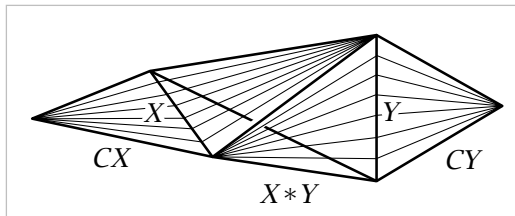
The simplest example of a stable splitting occurs for the torus $S^1 \times S^1$. Here the reduced suspension $\Sigma(S^1 \times S^1)$ is homotopy equivalent to $S^2 \vee S^2 \vee S^3$ since $\Sigma(S^1 \times S^1)$ is $S^2 \vee S^2$ with a 3-cell attached by the suspension of the attaching map of the 2-cell of the torus, but the latter attaching map is the commutator of the two inclusions $S^1 \hookrightarrow S^1$, and the suspension of this commutator is trivial since it lies in the abelian group $\pi_2(S^2 \vee S^2)$.

By an easy geometric argument we will prove more generally:

Proposition 4I.1. *If X and Y are CW complexes, then $\Sigma(X \times Y) \simeq \Sigma X \vee \Sigma Y \vee \Sigma(X \wedge Y)$.*

For example, $\Sigma(S^m \times S^n) \simeq S^{m+1} \vee S^{n+1} \vee S^{m+n+1}$. In view of the cup product structure on $H^*(S^m \times S^n)$ there can be no such splitting of $S^m \times S^n$ before suspension.

Proof: Consider the join $X * Y$ defined in Chapter 0, consisting of all line segments joining points in X to points in Y . For our present purposes it is convenient to use the reduced version of the join, obtained by collapsing to a point the line segment joining the basepoints $x_0 \in X$ and $y_0 \in Y$. We will still denote this reduced join by $X * Y$. Consider the space obtained from $X * Y$ by attaching reduced cones CX and CY to the copies of X and Y at the two ends of $X * Y$. If we collapse each of these cones to a point, we get the reduced suspension $\Sigma(X \times Y)$.



Since each cone is contractible, collapsing the cones gives a homotopy equivalence $X * Y \cup CX \cup CY \simeq \Sigma(X \times Y)$. Inside $X * Y$ there are also cones $x_0 * Y$ and $X * y_0$ intersecting in a point. Collapsing these cones converts $X * Y$ into $\Sigma(X \wedge Y)$ and $X * Y \cup CX \cup CY$ into $\Sigma(X \wedge Y) \vee \Sigma X \vee \Sigma Y$. $\square$

This result can be applied inductively to obtain splittings for suspensions of products of more than two spaces, using the fact that reduced suspension is smash product with S^1 , and smash product is associative and commutative. For example,

$$\Sigma(X \times Y \times Z) \simeq \Sigma X \vee \Sigma Y \vee \Sigma Z \vee \Sigma(X \wedge Y) \vee \Sigma(X \wedge Z) \vee \Sigma(Y \wedge Z) \vee \Sigma(X \wedge Y \wedge Z)$$

Our next example involves the reduced product $J(X)$ defined in §3.2. An interesting case is $J(S^n)$, which has a CW structure of the form $S^n \cup e^{2n} \cup e^{3n} \cup \dots$. All the cells e^{in} for $i > 1$ are attached nontrivially since $H^*(J(S^n); \mathbb{Q})$ is a polynomial ring $\mathbb{Q}[x]$ for n even and a tensor product $\mathbb{Q}[x] \otimes \Lambda_{\mathbb{Q}}[y]$ for n odd. However, after we suspend to $\Sigma J(S^n)$, it is a rather surprising fact that all the attaching maps become trivial:

Proposition 4I.2. $\Sigma J(S^n) \simeq S^{n+1} \vee S^{2n+1} \vee S^{3n+1} \vee \dots$. More generally, if X is a connected CW complex then $\Sigma J(X) \simeq \bigvee_n \Sigma X^{\wedge n}$ where $X^{\wedge n}$ denotes the smash product of n copies of X .

Proof: The space $J(X)$ is the union of an increasing sequence of subcomplexes $J_k(X)$ with $J_k(X)$ a quotient of the k -fold product $X^{\times k}$. The quotient $J_k(X)/J_{k-1}(X)$ is $X^{\wedge k}$. Thus we have maps

$$X^{\times k} \rightarrow J_k(X) \rightarrow X^{\wedge k} = J_k(X)/J_{k-1}(X)$$

By repeated application of the preceding proposition, $\Sigma X^{\wedge k}$ is a wedge summand of $\Sigma X^{\times k}$, up to homotopy equivalence. The proof shows moreover that there is a map $\Sigma X^{\wedge k} \rightarrow \Sigma X^{\times k}$ such that the composition $\Sigma X^{\wedge k} \rightarrow \Sigma X^{\times k} \rightarrow \Sigma X^{\wedge k}$ is homotopic to the identity. This composition factors as

$$\Sigma X^{\wedge k} \rightarrow \Sigma X^{\times k} \rightarrow \Sigma J_k(X) \rightarrow \Sigma X^{\wedge k}$$

so we obtain a map $s_k: \Sigma X^{\wedge k} \rightarrow \Sigma J_k(X)$ such that $\Sigma X^{\wedge k} \xrightarrow{s_k} \Sigma J_k(X) \rightarrow \Sigma X^{\wedge k}$ is homotopic to the identity.

The map s_k induces a splitting of the long exact sequence of homology groups for the pair $(\Sigma J_k(X), \Sigma J_{k-1}(X))$. Hence the map $i \vee s_k: \Sigma J_{k-1}(X) \vee \Sigma X^{\wedge k} \rightarrow \Sigma J_k(X)$ induces an isomorphism on homology, where i denotes the inclusion map. It follows by induction that the map $\bigvee_{k=1}^n s_k: \bigvee_{k=1}^n \Sigma X^{\wedge k} \rightarrow \Sigma J_n(X)$ induces an isomorphism on homology for all finite n . This implies the corresponding statement for $n = \infty$ since $X^{\wedge n}$ is $(n-1)$ -connected if X is connected. Thus we have a map $\bigvee_k \Sigma X^{\wedge k} \rightarrow \Sigma J(X)$ inducing an isomorphism on homology. By Whitehead's theorem this map is a homotopy equivalence since the spaces are simply-connected CW complexes. $\square$

For our final example the stable splitting will be constructed using the group structure on $\langle \Sigma X, Y \rangle$, the set of basepointed homotopy classes of maps $\Sigma X \rightarrow Y$.

Proposition 4I.3. For any prime power p^n the suspension $\Sigma K(\mathbb{Z}_{p^n}, 1)$ is homotopy equivalent to a wedge sum $X_1 \vee \dots \vee X_{p-1}$ where X_i is a CW complex having $\tilde{H}_*(X_i; \mathbb{Z})$ nonzero only in dimensions congruent to $2i \bmod 2p-2$.

This result is best possible in a strong sense: No matter how many times any one of the spaces X_i is suspended, it never becomes homotopy equivalent to a nontrivial wedge sum. This will be shown in Example 4L.3 by studying cohomology operations in $H^*(K(\mathbb{Z}_{p^n}, 1); \mathbb{Z}_p)$. There is also a somewhat simpler K-theoretic explanation for this; see [VBKT].

Proof: Let $K = K(\mathbb{Z}_{p^n}, 1)$. The multiplicative group of nonzero elements in the field $\mathbb{Z}_p$ is cyclic, so let the integer r represent a generator. By Proposition 1B.9 there is a map $f: K \rightarrow K$ inducing multiplication by r on $\pi_1(K)$. We will need to know that f induces multiplication by r^i on $H_{2i-1}(K; \mathbb{Z}) \approx \mathbb{Z}_{p^n}$, and this can be seen as follows. Via

the natural isomorphism $\pi_1(K) \approx H_1(K; \mathbb{Z})$ we know that f induces multiplication by r on $H_1(K; \mathbb{Z})$. Via the universal coefficient theorem, f also induces multiplication by r on $H^1(K; \mathbb{Z}_{p^n})$ and $H^2(K; \mathbb{Z}_{p^n})$. The cup product structure in $H^*(K; \mathbb{Z}_{p^n})$ computed in Examples 3.41 and 3E.2 then implies that f induces multiplication by r^i on $H^{2i-1}(K; \mathbb{Z}_{p^n})$, so the same is true for $H_{2i-1}(K; \mathbb{Z})$ by another application of the universal coefficient theorem.

For each integer $j \geq 0$ let $h_j: \Sigma K \rightarrow \Sigma K$ be the difference $\Sigma f - r^j \mathbb{1}$, so h_j induces multiplication by $r^i - r^j$ on $H_{2i}(\Sigma K; \mathbb{Z}) \approx \mathbb{Z}_{p^n}$. By the choice of r we know that p divides $r^i - r^j$ iff $i \equiv j \pmod{p-1}$. This means that the map induced by h_j on $\tilde{H}_{2i}(\Sigma K; \mathbb{Z})$ has nontrivial kernel iff $i \equiv j \pmod{p-1}$. Therefore the composition $m_i = h_1 \circ \cdots \circ h_{i-1} \circ h_{i+1} \circ \cdots \circ h_{p-1}$ induces an isomorphism on $\tilde{H}_*(\Sigma K; \mathbb{Z})$ in dimensions congruent to $2i \pmod{2p-2}$ and has a nontrivial kernel in other dimensions where the homology group is nonzero. When there is a nontrivial kernel, some power of m_i will induce the zero map since we are dealing with homomorphisms $\mathbb{Z}_{p^n} \rightarrow \mathbb{Z}_{p^n}$.

Let X_i be the mapping telescope of the sequence $\Sigma K \rightarrow \Sigma K \rightarrow \cdots$ where each map is m_i . Since homology commutes with direct limits, the inclusion of the first factor $\Sigma K \hookrightarrow X_i$ induces an isomorphism on $\tilde{H}_*$ in dimensions congruent to $2i \pmod{2p-2}$, and $\tilde{H}_*(X_i; \mathbb{Z}) = 0$ in all other dimensions. The sum of these inclusions is a map $\Sigma K \rightarrow X_1 \vee \cdots \vee X_{p-1}$ inducing an isomorphism on all homology groups. Since these complexes are simply-connected, the result follows by Whitehead's theorem. $\square$

The construction of the spaces X_i as mapping telescopes produces rather large spaces, with infinitely many cells in each dimension. However, by Proposition 4C.1 each X_i is homotopy equivalent to a CW complex with the minimum configuration of cells consistent with its homology, namely, a 0-cell and a k -cell for each k congruent to $2i$ or $2i+1 \pmod{2p-2}$.

Stable splittings of $K(G, 1)$'s for finite groups G have been much studied and are a complicated and subtle business. To take the simplest noncyclic example, Proposition 4I.1 implies that $\Sigma K(\mathbb{Z}_2 \times \mathbb{Z}_2, 1)$ splits as the wedge sum of two copies of $\Sigma K(\mathbb{Z}_2, 1)$ and $\Sigma(K(\mathbb{Z}_2, 1) \wedge K(\mathbb{Z}_2, 1))$, but the latter summand can be split further, according to a result in [Harris & Kuhn 1988] which says that for G the direct sum of k copies of $\mathbb{Z}_{p^n}$, $\Sigma K(G, 1)$ splits canonically as the wedge sum of pieces having exactly $p^k - 1$ distinct homotopy types. Some of these summands occur more than once, as we see in the case of $\mathbb{Z}_2 \times \mathbb{Z}_2$.

Exercises

1. If a connected CW complex X retracts onto a subcomplex A , show that $\Sigma X \simeq \Sigma A \vee \Sigma(X/A)$. [One approach: Show the map $\Sigma r + \Sigma q: \Sigma X \rightarrow \Sigma A \vee \Sigma(X/A)$ induces an isomorphism on homology, where $r: X \rightarrow A$ is the retraction and $q: X \rightarrow X/A$ is the quotient map.]